

Problem

When the input to a system is a unit step function, the response is $120 \cos 2tu(t)$. Obtain the transfer function of the system.

Step-by-step solution

Step 1 of 2

The response of a system for a unit step function is,

$$y(t) = 120 \cos 2tu(t)$$

Use the following transform pair to find the Laplace transform of the output function:

$$\cos atu(t) \xleftrightarrow{L} \frac{s}{s^2 + a^2}$$

The Laplace transform of the output function is,

$$\begin{aligned} Y(s) &= 120 \left(\frac{s}{s^2 + 2^2} \right) \\ &= \frac{120s}{s^2 + 4} \end{aligned}$$

The Laplace transform of the unit step function is,

$$X(s) = \frac{1}{s}$$

Step 2 of 2

Calculate the transfer function of the system.

$$H(s) = \frac{Y(s)}{X(s)}$$

Substitute the expressions of $X(s)$ and $Y(s)$.

$$\begin{aligned} H(s) &= \frac{\frac{120s}{s^2 + 4}}{\frac{1}{s}} \\ &= \frac{120s^2}{s^2 + 4} \end{aligned}$$

Hence, the transfer function of the system is $\boxed{\frac{120s^2}{s^2 + 4}}$.